

Advancing nature and biodiversity insights in the North Sea

Habitat sensitivity assessment in North Sea

Points of contact

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Repository Structure

- **data**
 - **raw_data:** the raw data integrated in the analysis (**Note:** original data name and structure were kept except when either name was not descriptive or similar data were put in same directory to simplify input directories)
 - **intermediate_data:** disaggregated processed data
 - **hex_data:** geopackage

Note for PC users: The code was written on a Mac so to run the scripts replace “/” in the pathnames for directories with two “\”.

Please contact Brian Free (brian.free@oceandata.earth) with any questions regarding the code.

Overview

Study region

The study defined a boundary box for the Greater North Sea with points of:

- southwest: -4.4454, 50.9954
- northwest: -4.4454 61.0170
- northeast: 12.0059, 61.0170
- southeast: 12.0059, 50.9954

Since the desired resolution for the analysis was at 500m, a hex grid at resolution 8 generated the hex grid using H3 indexes within the Greater North Sea. Marine Regions provided the boundary layer of the Greater North Sea (as defined by ICES) in the final analysis.

Data sources

Generic Data

Layer	Data Source	Data Name	Metadata	Notes
Study area	mregions2	Marine Regions	Greater North Sea	Defined by ICES Ecoregions
Platforms	Total Energies			
Species list	Natura 2000	Doggerbank		1 habitat, 7 species
Species list	Natura 2000	Sydlig Nordsø		0 habitats, 0 species
Species list	Natura 2000	Sydlig Nordsø		1 habitat, 3 species

Layer	Data Source	Data Name	Metadata	Notes
Species list	Natura 2000	Jyske Rev, Lillefiskerbanke		1 habitat, 0 species
Species list	Natura 2000	Sandbanker ud for Thyborøn		2 habitats, 1 species
Species list	Natura 2000	Sylter Aussenriff		2 habitat, 17 species

Nature Data

Layer	Data Source	Data Name	Metadata	Notes
Fish	Cesc Gordó-Vilaseca	Fish species richness	GitHub repository	
Cetaceans	MPA EU and provided by Silas C. Principe	Cetaceans species distribution models		GitHub, Shiny OBIS
Seabirds	MPA EU and provided by Silas C. Principe	Seabirds species distribution models		GitHub, Shiny OBIS
Conservation areas	Marine Mammals Protected Areas Task Force	Important Marine Mammal Areas		Need to request download
Protected areas	Protected Seas	Marine protected areas	Ocean Data Platform	Used the focused data set
Corals	Tong et al. (2023)	Cold water corals		Direct data download
Seagrass	WCMC	Seagrass	ISO	Ocean Data Platform, WCMC direct data download

Methods

Geographic constraint

Each nature data set got limited to the Greater North Sea boundary before performing any further data transformations.

Linear normalization

A linear function normalized species richness values for fish, cetaceans, and seabirds. Normalizing the minimum and maximum species richness to be between 0 and 1 allowed these data to get compared to the data sets that receive a discrete value for presence and absence (1 or 0).

The linear normalized function is: $(\text{value} - \text{minimum}) / (\text{maximum} - \text{minimum}) = \text{normalized score}$.

To avoid returning a zero for the normalized score (when the value is equal to the minimum), when the linear function returns 0.00, then a value of 0.01 got added. By adding value to the minimum value, it ensures that any species richness never would get treated as absence.

Overlapping features

When a hex grid received more than one value from a linear normalization, due to the shape incongruity, then the maximum value was kept. Selecting the maximum value ensured the study pursued an approach

that reflected a precautionary principle (read here for further context on the principle) since the nature index would represent the best case scenario for nature.

Habitats

Cold water corals The cold water corals data set contained values that range between 0 and 1000 for 10 species. Suitability classifications on coral presence are separated by steps of 200: very low (<200), low (200 - 400), moderate (400 - 600), high (600 - 800), and very high (800). This study only considered if a cold water coral was present, not the overall species richness. To determine if a cold water coral reef was present, the maximum score across the 10 species was returned and only values above 600 were kept.

Species

Cetaceans and seabirds Cetaceans considered in the analysis either had at least one observation in the Joint Cetacean Database Programme (for more information about the JCDP and ICES statistical surveys) or included in the Waggitt et al. (2020). The only species from Waggitt et al. (2020) that did not have at least one observation in the ICES statistical survey was *Physeter macrocephalus*.

These species included: - *Balaenoptera acutorostrata* - *Balaenoptera physalus* - *Delphinus delphis* - *Grampus griseus* - *Halichoerus grypus* - *Lagenorhynchus acutus* - *Lagenorhynchus albirostris* - *Orcinus orca* - *Phoca vitulina* - *Phocoena phocoena* - *Physeter macrocephalus* - *Stenella coeruleoalba* - *Tursiops truncatus*

Species compiled for seabirds came from Waggitt et al. (2020), ICES's European Seabirds at Seas, and a list provided by Total Energies. Species from the top 50 in ICES's ESAS supplemented the species already included in Waggitt et al. (2020) and the list provided by Total. The species included in the the lists are:

- *Alca torda*
- *Alle alle*
- *Aythya marila*
- *Bucephala clangula*
- *Cephus grylle*
- *Chlidonias niger*
- *Clangula hyemalis*
- *Fratercula arctica*
- *Fulmarus glacialis*
- *Gavia adamsii*
- *Gavia arctica*
- *Gavia immer*
- *Gavia stellata*
- *Gelochelidon nilotica*
- *Hydrobates leucorhous*
- *Hydrobates pelagicus*
- *Hydrocoloeus minutus*
- *Hydroprogne caspia*
- *Larus argentatus*
- *Larus canus*
- *Larus fuscus*
- *Larus glaucoides*
- *Larus hyperboreus*
- *Larus marinus*
- *Larus melanocephalus*
- *Larus michahellis*
- *Larus ridibundus*
- *Melanitta fusca*
- *Melanitta nigra*
- *Mergus merganser*

- *Mergus serrator*
- *Morus bassanus*
- *Phalacrocorax aristotelis*
- *Phalacrocorax carbo*
- *Phalaropus fulicarius*
- *Phalaropus lobatus*
- *Podiceps auritus*
- *Podiceps cristatus*
- *Podiceps grisegena*
- *Podiceps nigricollis*
- *Puffinus puffinus*
- *Rissa tridactyla*
- *Somateria mollissima*
- *Somateria spectabilis*
- *Stercorarius longicaudus*
- *Stercorarius parasiticus*
- *Stercorarius pomarinus*
- *Stercorarius skua*
- *Sterna hirundo*
- *Sterna paradisaea*
- *Sternula albifrons*
- *Thalasseus sandvicensis*
- *Uria aalge*
- *Xema sabini*

Species distribution models Species distribution models for the interested species came from the MPA Europe project on modeling ocean biodiversity. The study used species distribution model data from MPA Europe given they were produced at 5km resolution; comparatively, Waggitt et al. (2020) modeled their species at 10km. To maintain comparability between species, only species with modeled data by MPA Europe got included. Species were matched by their WoRMS Aphia identification. A marine mammal and two seabird species had different accepted species names in WoRMS than those used by ICES, Waggitt et al. (2020), and Total Energies. The three species were:

- *Lagenorhynchus acutus* (137100) – accepted name is *Leucopleurus acutus* (1571853)
- *Larus melanocephalus* (137147) – accepted name is *Ichthyæetus melanocephalus* (1584284)
- *Thalasseus sandvicensis* (413044) – accepted name is *Sterna sandvicensis* (137166)

MPA Europe did not possess a species distribution model for a species on the Total Energies list (*Hydrobates leucorhous*). The study thus did not include *Hydrobates leucorhous* in the final analysis.

Habitat designation MPA Europe’s species distribution models relied on presence-only data. The best modeled outputs were LASSO and maxent (maximum entropy). LASSO results were not available for download, so the priority were maxent along with ensemble outputs when the maxent models did not exist for a particular species. Most species have the maxent models, but five had ensemble models:

Cetaceans - *Phoca vitulina* (137084) - *Orcinus orca* (137102) - *Tursiops truncatus* (137111)

Seabirds - *Alle alle* 137129 - *Rissa tridactyla* 137156

Species distribution model (SDM) results were scored to reflect the match to likely habitat. Any location with a species distribution model value under 50 (quantitatively more probable than not) was classified as unlikely habitat and given a score 0.3. Species distribution model values between 50 and 75 got a score of 0.6 to reflect that these areas are probable habitat. Core habitat occurred in locations with values over 75 and they received a value of 1.

SDM value	Classification	Score
0 - 50	Unlikely habitat	0.3
50 - 75	Unlikely habitat	0.6
75+	Unlikely habitat	1.0

Normalization A summarized total species richness layer were created for each cetaceans and seabirds by combining all species values for each. The linear normalization was run on the summarized data products and any instance where multiple values existed for the same hex, the maximum normalized value was selected.

Fishes

A summarized fish species data layer came from a paper by Gordó-Vilaseca et al. (2024) that examined biomass distributions for a variety of species across the North Sea and Barents Sea. The data shared by Gordó-Vilaseca are the present day species richness data (as seen in the paper’s Figure 2(A)).

Normalization The Greater North Sea boundary limited the data to the study region before applying a linear normalization and rescaling the minimum species richness.

Scores

Layer	Score	Consideration
Fish	0 - 1	Linear normalization with 0.01 added to minimum
Cetaceans	0 - 1	Species list compiled from list provided by Total Energies, Waggitt et al. (2020), along with cetaceans that have at least one observation in the ICES statistical surveys for the North Sea
Seabirds	0 - 1	Species list compiled from list provided by Total Energies, Waggitt et al. (2020), along with data from European Seabirds at Seas and MPA EU
Protected areas	0, 1	Absence, Presence
Cold water corals	0, 1	Absence, Presence
Important Marine Mammal Areas	0, 1	Absence, Presence
Seagrass	0, 1	Absence, Presence

Submodel

Two submodels comprised the final nature score index. The fish, cetacean, and seabird data products were summarized in the mobile species submodel. The sessile submodel comprised of the other four datasets: protected areas, important marine mammal areas, cold water corals, and seagrass. Any “NA” data for a data set got reclassified as a 0 before calculating submodel scores. These two submodels then were combined to produce a final nature score. More weight was allotted to the sessile submodel since at least two of the layers are immovable, making them more reliably known. The analysis weighted the sessile submodel with 55% of the score. It did not receive greater share of weight as the seagrass and coral reef data sets remain to have a level of uncertainty concerning their verified locations.

Below are the three main equations for calculating submodel and final model scores:

$$[1] \text{ mobile_submodel_score} = \text{fish_value} + \text{cetacean_value} + \text{seabird_value}$$

[2] sessile_submodel_score = protected_area_value + imma_value + cold_water_coral_value + seagrass_value

[3] final_model_score = mobile_submodel_score * 0.45 + sessile_submodel_score * 0.55

Considerations / Limitations / Assumptions

Data components

A major limitation of the study was the inability to integrate more data sets for the nature index. The analysis desired to include more data sets and alternatives to the ones that got ingested, but had barriers for inclusion. For instance, while the UNEP-WCMC seagrass is an imperfect data set given that it was created as a global data set, and cannot accurately represent localized, high resolution areas, it was the best alternative available. A seagrass layer exists that has 250m resolution for the North Sea (report). The data were only for non-commercial use.

Two other data sets that would contribute to understanding bird species in the North Sea were limited by the inability to use them for commercial uses. BirdLife and SeaTrack contain species data with North Sea coverage. Along with these seabird data, if the analysis could expand build its own species distribution models, ICES's European Seas at Sea and its Joint Cetacean Data Programme data would provide years of presence data. Without building a separate species distribution models, an faster step to create a more comprehensive model is to include more species already modeled by the MPA Europe project. As of August 2025, the project produced species distribution models for 12,039 species (explore main results). If an alternative were found, the most appealing data product would be a species distribution model that integrated absence data. Presence-only models, while easier to generate, have more accurate outputs.

Finding alternative data to those included in the analysis could help address a major limitation with the analysis – the spatial resolution for the fish, cetacean, and seabird data layers. Downscaling the fish data set from 10km and the cetacean and seabird data sets from 5km generates numerous issues that generate inherent issues with evaluating the nature score with significant confidence. To improve the model outputs, data sets should have more aligned resolutions and ones that are closer to the desired output (in the case of this study, 500m). Additionally it is more preferable to not have a summarized data product for species richness that had to get used for the fish data layer. In a more ideal scenario, the species richness would get create in a more similar process to how this analysis generated the summarized data product for cetaceans and seabirds.

Analysis components

An assumption was made that the maximum score would get kept to reflect a higher nature score. Another major assumption was the inclusion of species even if it only had a single observation in the study region. Although the inclusion does allow for a more precautionary approach for representing nature, it may not allow for the most accurate reflection. This complicated how the species distribution models got scored, as potential distribution needed to get differentiated from “real” distribution. Species distribution model outputs of 1 should not get evaluated the same as one with 99. With more time, the analysis could undertake more nuance to define unlikely habitat, probable habitat, and core habitat.

Future

Explore how species interact with areas that are defined for protected areas in the North Sea – though should be aware of how the areas might get designation for a particular species due to nesting or another part of the life cycle that limits the time spent in that specific geography.